

Практикум С5

137 marks of the archive, eight techniques, twelve tasks. Everything the archive asks about vectors without a plane and without a vector product, from May 2021 to November 2025: points and lengths, the scalar product, the angle between two vectors, the equation of a line, the angle between two lines, where two lines meet — and whether they meet at all — and motion along a line.

О практикуме

Тема разрезана по действию, а не по разметке. `geometry.vectors` и `geometry.vectors_3d` делят вопросы не по тому, что с векторами делают: прямые и их пересечение лежат в обеих, плоскости и векторное произведение — тоже в обеих. С5 берёт всё, что делается без плоскости и без векторного произведения: 96 баллов из 182 `geometry.vectors` (0.527) и 52 из 216 `geometry.vectors_3d` (0.241). 148 баллов по корпусу, 137 рабочих — ноябрь 2023 года стоит дважды, пирамида Q1 и прямые Q12(a). 58 баллов Paper 1, 71 Paper 2, 8 Paper 3.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-c5-vectors

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

C5 — Vectors: points, lines and the angles between them

137 marks of the archive, eight techniques, twelve tasks. Everything the archive asks about vectors without a plane and without a vector product, from May 2021 to November 2025: points and lengths, the scalar product, the angle between two vectors, the equation of a line, the angle between two lines, where two lines meet — and whether they meet at all — and motion along a line.

Paper 1 and Paper 2 share it almost equally, 58 and 71 marks, and on both the work is the same: a system of two equations and a check of the third.

The one idea

A line is **a set of points**, and every question about it is a question about which points belong to it.

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$$

is one point **a** of the line and one step **b** along it. Any other point of the line and any multiple of the step describe **the same line**.

And what follows from it

the question asks	what it is
a vector equation of a line	any point on it, any direction along it
the angle between two lines	the angle between the directions, the acute one
the point of intersection	a point that belongs to both sets
<i>show that the lines are skew</i>	no common point, and the directions are not parallel

How the checks work

They do not know the answers. A line is handed to them **the way the question prints it**, and a point is handed **the conditions** that fix it:

```
L1 = cartesian((x + 1)/2, y, 3 - z)
verify_line('7a', q7a, L1)

D = unknown('D', 3)
verify_find('1a', q1a, D, [parallelogram(A,
B, C, D)])
```

The first is *"is your line the set of points $(x + 1)/2 = y = 3 - z$?"* — any point of it and any parallel direction pass. The second is *"does your D make $ABCD$ a parallelogram?"*: the check solves the condition itself.

When you are wrong the check says **how**:

what you wrote	what the check says
the point $(-1, 2, 0)$ for $\frac{x-1}{2} = \frac{y+2}{3} = z$	the point has the wrong signs
the position vector of B as the direction	not a step along the line
λ of one line put into the other	each line has its own parameter
the obtuse angle between the directions	between lines, take the acute one
the angle between $\overrightarrow{OB}$ and $\overrightarrow{OC}$	the angle at V is between the edges from V
<i>skew</i> for parallel lines	parallel lines are not skew

Order of work

level	what it means	tasks
	points and lengths, the scalar product	1-3
	angles, the equation of a line, intersection	4-7
	parallel, meeting or skew; motion along a line	8-12

Every task is a real past-paper question, cited.

79 of the 137 marks are on a calculator paper, and the calculator does little of the work. It extracts a square root, takes an inverse cosine, solves one equation for p . The equation of a line, the intersection and *skew* are the same system of equations on either paper.

```

import sys
sys.path.append('..')          # from
practicum/geometry to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
vec, line, cartesian, through, unknown

language('en')                # this
notebook is in English, and so are the checks

# vec(1, -4, 0) is a vector or a point;
dot(u, v), mag(v), distance(A, B), angle(u,
v).
# A line: line(vec(1, 2, -3) + s*vec(2, 3,
6)), through(A, B), cartesian((x - 1)/2, y, 3
- z).
# lam and mu are the Greek letters  $\lambda$  and  $\mu$ ; z
is the third coordinate.
# unknown('D', 3) is a point the question
asks for, and the checks find it themselves.

print('ready; sympy', sp.__version__)
A, B = vec(2, -1, 3), vec(6, 1, -1)
print('AB: ', B - A)
print('|AB|: ', mag(B - A))
print('the line AB: ', through(A, B))

```

Map of the eight techniques

#	technique	you recognise it by	it reduces to
1	points and lengths	<i>find the coordinates of D, find BV, the centre and the radius</i>	$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}, \mathbf{v} $
2	the scalar product	<i>perpendicular, a right angle, $\overrightarrow{OM} \cdot \overrightarrow{MC} = 0$</i>	$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = 0$
3	the angle between vectors	<i>the size of $\hat{BVC}$, the angle between $\mathbf{p}$ and $\mathbf{n}$</i>	$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} \mathbf{b} }$
4	the equation of a line	<i>find a vector equation of L_1, passes through A and B</i>	a point and a direction
5	the angle between lines	<i>the acute angle between the lines</i>	the directions only, $ \cos \theta $
6	the point of intersection	<i>find where L_1 and L_2 intersect</i>	two parameters, two equations, the third checked

#	technique	you recognise it by	it reduces to
7	parallel, meeting or skew	<i>do not intersect, show that the lines are skew, perpendicular and intersect</i>	directions first, then the system
8	motion along a line	<i>speed, bearing, the time between the arrivals, the angle of descent</i>	$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$

Techniques 1–3 are about vectors alone. Techniques 4–7 put a vector to work as the direction of a line: first the line itself, then two lines at once. Technique 8 is a line walked in time.

● Part 1. Points, lengths and the scalar product

Theory: from points to vectors

Take $A(2, -1, 3)$, $B(6, 1, -1)$ and $C(3, 4, 0)$.

A vector between two points is the end minus the start:

$$\begin{aligned}\overrightarrow{AB} &= \mathbf{b} - \mathbf{a} \\ &= \begin{pmatrix} 4 \\ 2 \\ -4 \end{pmatrix}, \quad |\overrightarrow{AB}| \\ &= \sqrt{16 + 4 + 16} \\ &= 6\end{aligned}$$

$\overrightarrow{BA}$ is the same vector the other way:
 $-\overrightarrow{AB}$.

The midpoint of $[AB]$ is the average of the ends, $(4, 0, 1)$ — not half of $\overrightarrow{AB}$, which is a step, not a place.

A parallelogram $ABCD$ has its vertices in order, so $\overrightarrow{AB} = \overrightarrow{DC}$: the fourth vertex is $\mathbf{d} = \mathbf{a} + \mathbf{c} - \mathbf{b} = (-1, 2, 4)$. The order

matters: $\mathbf{a} + \mathbf{b} - \mathbf{c}$ is also a parallelogram, but its vertices read $ABDC$.

Three points on one line give parallel vectors. $P(1, 0, 2)$, $Q(3, 1, 5)$, $R(r, 3, 11)$: $\overrightarrow{PR} = (r - 1, 3, 9)$ must be $3\overrightarrow{PQ} = (6, 3, 9)$, so $r = 7$.

A length and a direction. With $|\mathbf{a}| = 5$ and $|\mathbf{b}| = 10$, $|\mathbf{a} + \mathbf{b}|$ is largest when $\mathbf{b}$ points the same way as $\mathbf{a}$ (15) and smallest when it points the opposite way (5). In a *given that it is a minimum* question the answer is the vector $\mathbf{a} + \mathbf{b}$, not $\mathbf{b}$.

Task 1 ● — **May 2025 TZ3 Paper 1**
Q11(a), (b), 4 marks

The points $A(1, -4, 0)$, $B(-3, -6, 2)$, $C(-1, -2, 4)$ and D form a parallelogram, $ABCD$, where D is diagonally opposite B .

(a) Find the coordinates of D .

The diagonals of the parallelogram, $[AC]$ and $[BD]$, intersect at point E .

(b) Find the coordinates of E .

The check for (b) finds D itself, so a slip in (a) costs (a) only.

```
qla = ...          # D, as (x, y, z)
qlb = ...          # E

A, B, C = vec(1, -4, 0), vec(-3, -6, 2),
vec(-1, -2, 4)
D, E = unknown('D', 3), unknown('E', 3)

verify_find('la', qla, D, [parallelogram(A,
B, C, D)])
verify_find('lb', qlb, E, [parallelogram(A,
B, C, D), on(E, A, C), on(E, B, D)])
```

Task 2 — **May 2022 TZ1 Paper 2** **Q7, 9 marks**

Consider the vectors **a** and **b** such that $\mathbf{a} = \begin{pmatrix} 12 \\ -5 \end{pmatrix}$ and $|\mathbf{b}| = 15$.

(a) Find the possible range of values for $|\mathbf{a} + \mathbf{b}|$.

Consider the vector **p** such that $\mathbf{p} = \mathbf{a} + \mathbf{b}$.

(b) Given that $|\mathbf{a} + \mathbf{b}|$ is a minimum, find **p**.

Consider the vector **q** such that $\mathbf{q} = \begin{pmatrix} x \\ y \end{pmatrix}$, where $x, y \in \mathbb{R}^+$.

(c) Find $\mathbf{q}$ such that $|\mathbf{q}| = |\mathbf{b}|$ and $\mathbf{q}$ is perpendicular to $\mathbf{a}$.

*In (a) answer `Interval(lowest, highest)`.
The checks let $\mathbf{b}$ run round the whole circle of radius 15.*

```
q2a = ...          # Interval(lowest, highest)
q2b = ...          # p, as vec(...)
q2c = ...          # q, as vec(...)

theta = symbols('theta')
a = vec(12, -5)
b = 15*vec(cos(theta), sin(theta))      #
every b with |b| = 15
q = unknown('q', 2)

verify_extent('2a', q2a, mag(a + b), theta,
Interval(0, 2*pi))
verify_optimum('2b', q2b, a + b, theta,
Interval(0, 2*pi), 'min')
verify_find('2c', q2c, q, [perpendicular(q,
a), length(q, 15), q[0] > 0, q[1] > 0])
```

Theory: the scalar product

By components it is a number, not a vector:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3$$

For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (4, 5, -1)$ it is $8 - 5 - 3 = 0$: **zero means perpendicular**,

and that is the whole content of *perpendicular* and *a right angle* in a question.

A perpendicular vector of a given length. Perpendicular to $(8, 15)$ is the direction $(15, -8)$ — swap and change one sign. Its length is 17; for a length of 34 double it: $(30, -16)$, or $(-30, 16)$ if the question wants the other one. Two conditions, two unknowns, and the question's restriction picks the sign.

With letters for vectors expand like brackets, remembering $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$ and $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$:

$$(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = |\mathbf{a}|^2 - |\mathbf{b}|^2$$

— so the diagonals of a rhombus, where $|\mathbf{a}| = |\mathbf{b}|$, are perpendicular.

A right angle at a vertex uses the vectors **from that vertex**. For $C(1, 1)$, $A(4, 5)$, $B(5, -2)$: $\overrightarrow{CA} = (3, 4)$, $\overrightarrow{CB} = (4, -3)$, and $12 - 12 = 0$.

Task 3 ● — **May 2023 TZ2 Paper 1**
Q9(a), (b), 5 marks

The following diagram shows parallelogram $OABC$ with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$ and $|\mathbf{c}| = 2|\mathbf{a}|$, where $|\mathbf{a}| \neq 0$. (In the diagram O and C are the bottom side, A and B the top side, and the angle $\widehat{OMC}$ is marked as a right angle.)

The angle between $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is θ , where $0 < \theta < \pi$.

Point M is on $[AB]$ such that $\overrightarrow{AM} = k\overrightarrow{AB}$, where $0 \leq k \leq 1$ and $\overrightarrow{OM} \cdot \overrightarrow{MC} = 0$.

(a) Express $\overrightarrow{OM}$ and $\overrightarrow{MC}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

(b) Hence, use a vector method to show that $|\mathbf{a}|^2(1 - 2k)(2 \cos \theta - (1 - 2k)) = 0$.

For (b) enter $\overrightarrow{OM} \cdot \overrightarrow{MC}$ in terms of $|\mathbf{a}|$ (1a), θ and k — the line before you factorise. The checks draw the parallelogram for several $|\mathbf{a}|$, θ and k . Part (c) is in C3.

```

la, theta = symbols('|a| theta',
positive=True)
a = la*vec(1, 0)
c = 2*la*vec(cos(theta), sin(theta))    # |c|
= 2|a|, and the angle between them is theta

q3a_OM = ...      # OM in terms of a and c
q3a_MC = ...      # MC in terms of a and c
q3b = ...         # OM · MC in terms of la,
theta and k

O = vec(0, 0)
B, M = unknown('B', 2), unknown('M', 2)
figure = [parallelogram(O, a, B, c,
names='OABC'), Eq(M - a, k*(B - a))]

verify_find('3a OM', q3a_OM, M - O, figure)
verify_find('3a MC', q3a_MC, c - M, figure)
verify_find('3b', q3b, dot(M - O, c - M),
figure)

```

Part 2. Angles, lines and where they meet

Theory: the angle between two vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}, \quad 0 \leq \theta \leq \pi$$

The angle at a vertex uses the edges from that vertex. For the triangle $P(1, 0, 0)$, $Q(0, 2, 0)$, $R(0, 0, 3)$ the angle at P is between $\overrightarrow{PQ} = (-1, 2, 0)$ and $\overrightarrow{PR} = (-1, 0, 3)$:

$$\begin{aligned}\cos Q\hat{P}R &= \frac{1}{\sqrt{5}\sqrt{10}} \\ &= 0.141, \quad Q\hat{P}R \\ &= 81.9^\circ\end{aligned}$$

The position vectors $\overrightarrow{OQ}$ and $\overrightarrow{OR}$ are perpendicular, and 90° is the angle at the origin — not at P .

Divide by both lengths. $\mathbf{a} \cdot \mathbf{b}$ alone is not $\cos \theta$; it is $\cos \theta$ times two lengths.

A letter inside a vector turns the formula into an equation. When two

angles are equal, set the **cosines** equal. Squaring both sides is tempting and dangerous: it also solves "the cosines are equal in size and opposite in sign", and that extra root answers a different question.

Units. Paper 2 accepts radians or degrees unless the question names one; three significant figures either way.

Task 4 ● — **November 2023 TZ1**
Paper 2 Q1, 6 marks

The following diagram shows a pyramid with vertex V and rectangular base $OABC$.

Point B has coordinates $(6, 8, 0)$, point C has coordinates $(6, 0, 0)$ and point V has coordinates $(3, 4, 9)$.

(a) Find BV .

(b) Find the size of $\hat{BVC}$.

```
q4a = ...          # BV
q4b = ...          # the angle BVC
```

```
B, C, V = vec(6, 8, 0), vec(6, 0, 0), vec(3,
4, 9)
```

```
verify_find('4a', q4a, distance(B, V))
verify_angle('4b', q4b, B, V, C)
```

Task 5 — *May 2025 TZ3 Paper 2* **Q6, 8 marks**

Consider the vectors $\mathbf{a} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -8 \\ 9 \end{pmatrix}$
and $\mathbf{c} = \begin{pmatrix} p \\ -6 \end{pmatrix}$, where $p \in \mathbb{R}$.

(a) Find an expression, in terms of p , for
(i) $\mathbf{a} \cdot \mathbf{c}$; **(ii)** $\mathbf{b} \cdot \mathbf{c}$.

The angle between $\mathbf{a}$ and $\mathbf{c}$ is equal to the
angle between $\mathbf{b}$ and $\mathbf{c}$.

(b) Find the value of p .

```

p = symbols('p')

q5a_i = ...      # a · c in terms of p
q5a_ii = ...     # b · c in terms of p
q5b = ...        # p

a, b, c = vec(-5, 7), vec(-8, 9), vec(p, -6)

verify_find('5a(i)', q5a_i, dot(a, c))
verify_find('5a(ii)', q5a_ii, dot(b, c))
verify_find('5b', q5b, p, [Eq(angle(a, c),
angle(b, c))])

```

Theory: the equation of a line

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$$

— a point $\mathbf{a}$ on the line and a direction $\mathbf{b}$ along it.

Through two points $A(2, 0, -1)$ and $B(5, 1, 1)$: the direction is $\overrightarrow{AB} = (3, 1, 2)$, the point is either of them:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

The position vector $(5, 1, 1)$ is a place on the line, not a step along it.

From the Cartesian form set every part equal to λ :

$$\frac{x+3}{2} = \frac{y-1}{5} = 4-z = \lambda$$

$$\implies$$

$$x = -3 + 2\lambda,$$

$$y = 1 + 5\lambda,$$

$$z = 4 - \lambda$$

so the point is $(-3, 1, 4)$ — **the signs flip** — and the direction is $(2, 5, -1)$: in $4 - z$ the coefficient of z is -1 .

A point on a line gives one value of the parameter. Is $(8, 2, 3)$ on the line through A and B ? $2 + 3\lambda = 8$ gives $\lambda = 2$, and the other two components agree: $0 + 2 = 2$, $-1 + 4 = 3$. Yes, at $\lambda = 2$.

Write "r =". The markscheme gives A0 for the vector without it.

Task 6 — **May 2025 TZ2 Paper 1 Q2, 5 marks**

The line L_1 is defined by the Cartesian equation $\frac{x-1}{2} = \frac{y+2}{3} = z$.

(a) Find a vector equation of L_1 .

A second line L_2 is defined by the vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ 4 \\ -8 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$, where $t \in \mathbb{R}$.

(b) Find the coordinates of the point where L_1 and L_2 intersect.

Enter a line as `vec(point) + lam*vec(direction)`.

```
q6a = ...           # a vector equation of L1
q6b = ...           # the point of intersection

L1 = cartesian((x - 1)/2, (y + 2)/3, z)
L2 = line(vec(0, 4, -8) + t*vec(1, 0, 2))

verify_line('6a', q6a, L1)
verify_meet('6b', q6b, L1, L2)
```

Theory: the angle between two lines

The angle between two lines is the angle between **their directions** — the points on them do not enter at all — and it is the **acute** one:

$$\cos \theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1| |\mathbf{d}_2|}$$

For directions $(1, 2, 2)$ and $(2, -1, -2)$: $\mathbf{d}_1 \cdot \mathbf{d}_2 = -4$ and both lengths are 3. The formula without the modulus gives 116.4° — the angle between the arrows; the lines meet at $180^\circ - 116.4^\circ = 63.6^\circ$.

A letter in a direction. The acute angle between $(1, 1, 0)$ and $(0, 1, c)$ is 60° :

$$\begin{aligned}\frac{1}{\sqrt{2}\sqrt{1+c^2}} &= \frac{1}{2} \\ \implies \\ 1+c^2 &= 2 \\ \implies \\ c &= \pm 1\end{aligned}$$

The modulus becomes a square, and **both** roots are answers: the question asked for an acute angle, and each of them gives one.

Theory: where two lines meet

Give each line its own parameter. Even if the question calls both t , the point of intersection is reached at different values.

$L_1 : \mathbf{r} = (1, 2, 0) + \lambda(1, -1, 2)$ and $L_2 : \mathbf{r} = (4, 0, 7) + \mu(0, 1, 1)$.

1. **Equate the components** — three equations, two unknowns: $1 + \lambda = 4$, $2 - \lambda = \mu$, $2\lambda = 7 + \mu$.
2. **Solve two of them:** $\lambda = 3$, $\mu = -1$.
3. **Check the third:** $2 \cdot 3 = 6$ and $7 + (-1) = 6$ — it holds, so the lines meet.
4. **Substitute into the right line:** $\lambda = 3$ into L_1 gives $(4, -1, 6)$.

Putting $\lambda = 3$ into L_2 gives $(4, 3, 10)$ — a point, but not on L_1 .

A letter in a direction makes the parameter a fraction in that letter; the value that makes its denominator zero is the one where the lines have no unique point.

Task 7 — **May 2021 TZ1 Paper 1 Q11, 19 marks**

Consider the line L_1 defined by the Cartesian equation $\frac{x+1}{2} = y = 3-z$.

(a) (i) Show that the point $(-1, 0, 3)$ lies on L_1 . **(ii)** Find a vector equation of L_1 .

Consider a second line L_2 defined by the vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}$, where $t \in \mathbb{R}$ and $a \in \mathbb{R}$.

(b) Find the possible values of a when the acute angle between L_1 and L_2 is 45° .

It is given that the lines L_1 and L_2 have a unique point of intersection, A , when $a \neq k$.

(c) Find the value of k , and find the coordinates of the point A in terms of a .

Paper 1: (b) wants exact values, all of them. The check for (c) tries several a .

```

a = symbols('a')

q7a = ...          # a vector equation of L1
q7b = [...]        # all possible values of a,
exact
q7c_k = ...        # k
q7c_A = ...        # A in terms of a

L1 = cartesian((x + 1)/2, y, 3 - z)
L2 = line(vec(0, 1, 2) + t*vec(a, 1, -1), t)

verify_line('7a', q7a, L1)
verify_find('7b', q7b, a, [Eq(angle(L1, L2),
pi/4)], exact=True)
verify_find('7c k', q7c_k, a,
[no_unique_meet(L1, L2)])
verify_meet('7c A', q7c_A, L1, L2)

```

● Part 3. Parallel, meeting or skew — and motion along a line

Theory: three ways two lines can sit

In a plane two lines either meet or are parallel. In space there is a third way: **skew** — not parallel, and still no common point.

Look at the directions first. $(1, -1, 2)$ and $(-2, 2, -4)$ are multiples: the lines are parallel (or the same line, if a point of one lies on the other), and there is nothing to solve.

Not parallel — solve two components, test the third. $L_1 : \mathbf{r} = (1, 2, 0) + \lambda(1, -1, 2)$ and $L_3 : \mathbf{r} = (2, 0, 1) + \mu(1, 1, 0)$:

$$1 + \lambda = 2 + \mu,$$

$$2 - \lambda = \mu$$

$$\implies$$

$$\lambda = \frac{3}{2},$$

$$\mu = \frac{1}{2}$$

and the third component: $2\lambda = 3$ but $1 = 1$.
It fails: no common point.

Skew needs both reasons. The markscheme gives R1 for *not parallel* and R1 for the contradiction. Parallel lines never meet either — and they are not skew.

Conditions with letters. *Perpendicular* is $\mathbf{d}_1 \cdot \mathbf{d}_2 = 0$; *intersect* is "the system of components has a solution". Write both and solve them together: the scalar product usually gives one letter at once, and the components give the rest.

Task 8 ● — **November 2025 TZ3**
Paper 1 Q10(a)-(c), 8 marks

The point $P(-1, 1, -13)$ lies on the line L_1 .

The line L_1 has a direction vector $\begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$.

(a) Write down a vector equation for L_1 in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

(b) Find a vector equation for line L_2 in the form $\mathbf{s} = \mathbf{c} + \mu\mathbf{d}$, given that L_2 passes

through the points $A(2, -4, 2)$ and $B(7, -6, 1)$.

(c) Show that L_1 and L_2 are skew.

*For (c) name the relation — 'parallel', 'intersecting' or 'skew' — and enter $[\lambda, \mu]$ found from two of the component equations of **your** lines from (a) and (b). The check tells you what the third component does.*

```
q8a = ...          # L1: vec(...) +  
lam*vec(...)  
q8b = ...          # L2: vec(...) + mu*vec(...)  
q8c = ...          # 'parallel', 'intersecting'  
or 'skew'  
q8c_pair = [...] # [lam, mu] from two of the  
component equations  
  
point_P = vec(-1, 1, -13)  
A, B = vec(2, -4, 2), vec(7, -6, 1)  
L1, L2 = line(point_P, vec(7, 1, 2)),  
through(A, B)  
  
verify_line('8a', q8a, L1)  
verify_line('8b', q8b, L2)  
verify_relation('8c', q8c, L1, L2)  
verify_pair('8c pair', q8c_pair, q8a, q8b)
```

Task 9 ● — *May 2025 TZ1 Paper 1*
Q6, 6 marks

The line L_1 has vector equation $\mathbf{r} = 4\mathbf{i} - \mathbf{k} + \lambda(a\mathbf{j} + \mathbf{k})$, where $a, \lambda \in \mathbb{R}$.

The line L_2 has vector equation $\mathbf{r} = \mathbf{i} - b\mathbf{k} + \mu(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$, where $b, \mu \in \mathbb{R}$.

The lines L_1 and L_2 are perpendicular and intersect at a unique point.

Find the value of a and the value of b .

```
a, b = symbols('a b')

q9 = [...]          # [a, b]

L1 = line(vec(4, 0, -1) + lam*vec(0, a, 1),
lam)
L2 = line(vec(1, 0, -b) + mu*vec(1, 2, 3),
mu)

verify_find('9', q9, [a, b],
[perpendicular(L1, L2), meet(L1, L2)])
```

Task 10 ● — *November 2022 Paper 2*
Q12(a)-(c), 13 marks

Consider the points $A(1, 2, 3)$, $B(k, -2, 1)$ and $C(5, 0, 2)$, where $k \in \mathbb{R}$.

(a) Write down $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

(b) Given that the points A , B and C lie on a straight line, show that $k = 9$.

(c) For $k = 9$, let L_1 be the line passing through A , B and C .

(i) Find a vector equation of the line L_1 .

(ii) Line L_2 has the equation $\frac{x-1}{2} = \frac{y}{3} = 1-z$. Show that the lines L_1 and L_2 are skew.

For (b) enter the value of k that the collinearity gives. For (c)(ii) the parameter of L_2 is the common value μ of its three parts.

```

q10a_AB = ...          # AB in terms of k
q10a_AC = ...          # AC
q10b = ...             # k
q10ci = ...            # L1: vec(...) +
lam*vec(...)
q10cii = ...           # 'parallel',
'intersecting' or 'skew'
q10cii_pair = [...]    # [lam, mu] from two of
the component equations

A, B, C = vec(1, 2, 3), vec(k, -2, 1), vec(5,
0, 2)
L2 = cartesian((x - 1)/2, y/3, 1 - z)

verify_find('10a AB', q10a_AB, B - A)
verify_find('10a AC', q10a_AC, C - A)
verify_find('10b', q10b, k, [parallel(B - A,
C - A)])
verify_line('10c(i)', q10ci, through(A, C))
verify_relation('10c(ii)', q10cii, through(A,
C), L2)
verify_pair('10c(ii) pair', q10cii_pair,
q10ci, L2)

```

Theory: a line walked in time

$$\mathbf{r} = \mathbf{r}_0 + t \mathbf{v}$$

is where a ship or an aircraft is at time t : it starts at $\mathbf{r}_0$ and moves by $\mathbf{v}$ every unit of time.

Speed is $|\mathbf{v}|$ — everything that multiplies t , numbers in front included. A ship at

$\mathbf{r} = (2, -3) + t(3, 4)$ km moves at 5 km/h; at $\mathbf{r} = (2, -3) + 2t(3, 4)$ it moves at 10. The position vector $(2, -3)$ says nothing about speed.

A bearing is read from the horizontal components, east and north, measured **from north, clockwise**, in three figures. The ship goes 3 east and 4 north each hour: $\arctan \frac{3}{4} = 36.9^\circ$ from north, so 037° — not 053°, which is the angle from east.

Two paths that cross are two lines. Where they cross is technique 6 with **two times**, t_1 and t_2 ; setting $\mathbf{r}_A(t) = \mathbf{r}_B(t)$ with one t asks whether they collide. The time between their arrivals at the crossing is $|t_1 - t_2|$, in the units of t .

Climbing or descending. A drone with velocity $(6, -8, 0)$ m/s starts to descend at 3 m/s with its horizontal velocity unchanged: its velocity is $(6, -8, -3)$. The angle with the horizontal compares the vertical part with the horizontal speed 10: $\arctan \frac{3}{10} = 16.7^\circ$.

Task 11 ● — May 2022 TZ2 Paper 2
Q11(a)-(d), 15 marks

Two airplanes, A and B , have position vectors with respect to an origin O given respectively by

$$\mathbf{r}_A = \begin{pmatrix} 19 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} -6 \\ 2 \\ 4 \end{pmatrix}$$

$$\mathbf{r}_B = \begin{pmatrix} 1 \\ 0 \\ 12 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$$

where t represents the time in minutes and $0 \leq t \leq 2.5$.

Entries in each column vector give the displacement east of O , the displacement north of O and the distance above sea level, all measured in kilometres.

(a) Find the three-figure bearing on which airplane B is travelling.

(b) Show that airplane A travels at a greater speed than airplane B .

(c) Find the acute angle between the two airplanes' lines of flight. Give your

answer in degrees.

The two airplanes' lines of flight cross at point P .

(d) (i) Find the coordinates of P . **(ii)** Determine the length of time between the first airplane arriving at P and the second airplane arriving at P .

In (a) enter the bearing as a string of three figures. For (b) enter both speeds. Part (e), the minimum distance, is optimisation — E9.

```

q11a = ...      # the bearing of B, a string
of three figures
q11b_A = ...    # the speed of A, km per
minute
q11b_B = ...    # the speed of B
q11c = ...      # the acute angle, degrees
q11d_i = ...    # the point where the paths
cross
q11d_ii = ...   # the time between the
arrivals, minutes

rA = vec(19, -1, 1) + t*vec(-6, 2, 4)
rB = vec(1, 0, 12) + t*vec(4, 2, -2)
t1, t2, crossing = unknown('t1'),
unknown('t2'), unknown('crossing', 3)

verify_bearing('11a', q11a, velocity(rB))
verify_speed('11b A', q11b_A, rA)
verify_speed('11b B', q11b_B, rB)
verify_angle('11c', q11c, line(rA, t),
line(rB, t), deg=True)
verify_meet('11d(i)', q11d_i, line(rA, t),
line(rB, t))
verify_find('11d(ii)', q11d_ii, Abs(t1 - t2),
[Eq(rA.subs(t, t1), crossing), Eq(rB.subs(t,
t2), crossing)])

```

Task 12 — *May 2025 TZ1 Paper 2* **Q7, 5 marks**

At 09:00 a helicopter is located at a point $(10, 3, 0.5)$ relative to a point O on horizontal ground. The x -direction is due

east, the y -direction is due north and the z -direction is vertically upwards.

All distances are measured in kilometres.

The helicopter is flying at a constant height.

The helicopter's position relative to the point O is given by $\mathbf{r} = \begin{pmatrix} 10 \\ 3 \\ 0.5 \end{pmatrix} + 4t \begin{pmatrix} 10 \\ -25 \\ 0 \end{pmatrix}$, where t represents the time in hours since 09:00.

(a) Find the speed of the helicopter.

At 10:00 the helicopter begins to descend.

During descent the helicopter's vertical height decreases at a constant rate of 16 km h^{-1} and its horizontal velocity remains unchanged.

The angle of descent, β , is defined as the angle between the helicopter's direction of travel and the horizontal.

(b) Find β , giving your answer in degrees.

```
q12a = ...          # the speed, km/h
q12b = ...          # beta, degrees

r = vec(10, 3, 0.5) + 4*t*vec(10, -25, 0)
descending = velocity(r) + vec(0, 0, -16)
# the horizontal velocity unchanged, the
height falls 16 km/h

verify_speed('12a', q12a, r)
verify_angle('12b', q12b, descending,
'horizontal', deg=True)
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
points	points and lengths: a vector between points, a midpoint, a vertex
scalar	the scalar product: perpendicular, a right angle
angle	the angle between two vectors
line	the equation of a line
line_angle	the angle between two lines
meet	the point where two lines meet
skew	parallel, intersecting or skew; conditions on letters
motion	speed, bearing and time along a line

1. Find the coordinates of the midpoint of $[PQ]$, where $P(2, 5, -1)$ and $Q(4, -3, 7)$.
2. $\mathbf{u} = (3, -2, 1)$ and $\mathbf{v} = (p, 4, 2)$ are perpendicular. Find p .
3. Find the angle between the vectors $(1, 2, 2)$ and $(2, 0, -1)$.
4. A line passes through $(0, 3, -2)$ and $(4, 1, 5)$. Write down its vector equation.
5. Find the acute angle between $\mathbf{r} = (1, 0, 1) + \lambda(2, 1, 2)$ and $\mathbf{r} = (0, 5, 0) + \mu(1, -2, 2)$.

6. Find the point of intersection of $\mathbf{r} = (1, 1, 0) + \lambda(1, 0, 2)$ and $\mathbf{r} = (3, 2, 5) + \mu(0, -1, 1)$.
7. Determine whether $\mathbf{r} = (2, 0, 1) + \lambda(1, 3, -1)$ and $\mathbf{r} = (0, 1, 4) + \mu(2, 1, 1)$ are parallel, intersecting or skew.
8. A boat's position at time t hours is $\mathbf{r} = (4, -1) + t(5, 12)$ km. Find its speed.
9. $ABCD$ is a parallelogram with $A(1, 1)$, $B(4, 2)$ and $C(6, 5)$. Find D .
10. Show that the triangle with vertices $(0, 0, 0)$, $(2, 1, 2)$ and $(1, -2, 0)$ has a right angle.
1. Find a vector equation of the line $\frac{x-3}{4} = \frac{y+1}{-2} = \frac{z}{5}$.
2. An aircraft flies with velocity $(300, 400, 0)$ km/h, east and north. Find the bearing of its course.

```

answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: 'c54a061dfcc1', 2:
'da3a456ba70b', 3: '49387897bee2', 4:
'38a9c1e72158', 5: 'e26736118709', 6:
'8ccb033c0e48', 7: 'fd1478e5fc36', 8:
'305238273bb0', 9: 'c54a061dfcc1', 10:
'da3a456ba70b', 11: '38a9c1e72158', 12:
'305238273bb0'})

```

On the clock — *November 2023 TZ1 Paper 2 Q12(a), 5 marks*

Five marks, seven minutes. Calculator allowed, no hints.

Line L is given by the vector equation

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + s \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \text{ where } s \in \mathbb{R}.$$

Line M is given by the vector equation

$$\mathbf{r}_2 = \begin{pmatrix} 9 \\ 9 \\ 11 \end{pmatrix} + t \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \text{ where } t \in \mathbb{R}.$$

(a) Show that lines L and M intersect at a point A and find the position vector of A .

Attempt log

date	time	result

```
qt_pair = [...] # [s, t] from two of the
component equations
qt = ...        # the position vector of A

s = symbols('s')
L = line(vec(1, 2, -3) + s*vec(2, 3, 6), s)
M = line(vec(9, 9, 11) + t*vec(4, 1, 2), t)

verify_pair('timer [s, t]', qt_pair, L, M)
verify_meet('timer A', qt, L, M)
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a) $ABCD$ in order means $\overrightarrow{BC} = \overrightarrow{AD}$:
 $\overrightarrow{BC} = (2, 4, 2)$, so $D = A + (2, 4, 2) = \boxed{(3, 0, 2)}$.

1 (b) The diagonals of a parallelogram bisect each other: E is the midpoint of $[AC]$, $(\frac{1-1}{2}, \frac{-4-2}{2}, \frac{0+4}{2}) = \boxed{(0, -3, 2)}$.

2 (a) $|\mathbf{a}| = \sqrt{144 + 25} = 13$. The longest $\mathbf{a} + \mathbf{b}$ has $\mathbf{b}$ along $\mathbf{a}$, $13 + 15 = 28$; the shortest has it against $\mathbf{a}$, $15 - 13 = 2$:

$$\boxed{2 \leq |\mathbf{a} + \mathbf{b}| \leq 28}$$

2 (b) At the minimum $\mathbf{b} = -\frac{15}{13}\mathbf{a}$, so $\mathbf{p} = \mathbf{a} + \mathbf{b} = -\frac{2}{13}\mathbf{a} = \boxed{\begin{pmatrix} -24/13 \\ 10/13 \end{pmatrix}} = \begin{pmatrix} -1.85 \\ 0.769 \end{pmatrix}$.

2 (c) Perpendicular to $(12, -5)$ is the direction $(5, 12)$, of length 13; with $x, y > 0$

and length 15:

$$\mathbf{q} = \frac{15}{13} \begin{pmatrix} 5 \\ 12 \end{pmatrix} = \boxed{\begin{pmatrix} 75/13 \\ 180/13 \end{pmatrix}} = \begin{pmatrix} 5.77 \\ 13.8 \end{pmatrix}$$

3 (a) $\overrightarrow{AB} = \overrightarrow{OC} = \mathbf{c}$, so $\overrightarrow{OM} = \mathbf{a} + k\mathbf{c}$ and $\overrightarrow{MC} = \mathbf{c} - (\mathbf{a} + k\mathbf{c}) = \boxed{(1-k)\mathbf{c} - \mathbf{a}}$.

3 (b) Expanding, $\overrightarrow{OM} \cdot \overrightarrow{MC} = (1-2k)\mathbf{a} \cdot \mathbf{c} - |\mathbf{a}|^2 + k(1-k)|\mathbf{c}|^2$. With $|\mathbf{c}|^2 = 4|\mathbf{a}|^2$ and $\mathbf{a} \cdot \mathbf{c} = 2|\mathbf{a}|^2 \cos \theta$:

$$\boxed{2(1-2k)|\mathbf{a}|^2 \cos \theta - |\mathbf{a}|^2 + 4k(1-k)|\mathbf{a}|^2} \\ = |\mathbf{a}|^2(2(1-2k) \cos \theta - (1-2k)^2)$$

and this is $|\mathbf{a}|^2(1-2k)(2 \cos \theta - (1-2k)) = 0$.

4 (a) $BV = \sqrt{3^2 + 4^2 + 9^2} = \sqrt{106} = 10.2956 \dots = \boxed{10.3}$

4 (b) From V : $\overrightarrow{VB} = (3, 4, -9)$, $\overrightarrow{VC} = (3, -4, -9)$, and $\overrightarrow{VB} \cdot \overrightarrow{VC} = 9 - 16 + 81 = 74$:

$$\begin{aligned}
 \cos B\hat{V}C &= \frac{74}{\sqrt{106}\sqrt{106}} \\
 &= \frac{74}{106}, \quad B\hat{V}C \\
 &= 0.798037\dots \\
 &= \boxed{0.798} \quad (45.7^\circ)
 \end{aligned}$$

5 (a) $\mathbf{a} \cdot \mathbf{c} = -5p - 42$ and $\mathbf{b} \cdot \mathbf{c} = -8p - 54$.

5 (b) $|\mathbf{a}| = \sqrt{74}$, $|\mathbf{b}| = \sqrt{145}$, and $|\mathbf{c}|$ cancels:

$$\begin{aligned}
 \frac{-5p - 42}{\sqrt{74}} &= \frac{-8p - 54}{\sqrt{145}} \\
 &\implies \\
 p &= 4.78727\dots = \boxed{4.79}
 \end{aligned}$$

6 (a) $\frac{x-1}{2} = \frac{y+2}{3} = z = \lambda$ gives $x = 1 + 2\lambda$, $y = -2 + 3\lambda$, $z = \lambda$:

$$\boxed{\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}}$$

6 (b) $1 + 2\lambda = t$, $-2 + 3\lambda = 4$, $\lambda = -8 + 2t$: the second gives $\lambda = 2$, the first $t = 5$, and

the third holds, $2 = -8 + 10$. So $\boxed{(5, 4, 2)}$.

7 (a) (i) $\frac{-1+1}{2} = 0 = 3 - 3.$ **(ii)**

$$\boxed{\mathbf{r} = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}} \quad \text{— from } x = -1 + 2\lambda,$$

$$y = \lambda, \quad z = 3 - \lambda.$$

7 (b) $(2, 1, -1) \cdot (a, 1, -1) = 2a + 2,$ the lengths are $\sqrt{6}$ and $\sqrt{a^2 + 2}$:

$$\begin{aligned} \frac{|2a + 2|}{\sqrt{6}\sqrt{a^2 + 2}} &= \frac{1}{\sqrt{2}} \\ &\Rightarrow 4a^2 + 8a + 4 \\ &= 3a^2 + 6 \\ &\Rightarrow a^2 + 8a - 2 \\ &= 0 \end{aligned}$$

$$\Rightarrow \boxed{a = -4 \pm 3\sqrt{2}}$$

7 (c) $-1 + 2\lambda = at, \quad \lambda = 1 + t, \quad 3 - \lambda = 2 - t.$ The last two agree, and the first becomes $1 + 2t = at,$ so $t = \frac{1}{a-2}$: no solution when $a = 2,$ $\boxed{k = 2}$. Then

$$A = \left(\frac{a}{a-2}, 1 + \frac{1}{a-2}, 2 - \frac{1}{a-2} \right)$$

$$= \boxed{\left(\frac{a}{a-2}, \frac{a-1}{a-2}, \frac{2a-5}{a-2} \right)}$$

8 (a) $\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$

8 (b) $\overrightarrow{AB} = (5, -2, -1):$

$$\mathbf{s} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix}$$

8 (c) $\frac{5}{7} \neq \frac{-2}{1}$: not parallel. The first two components: $-1 + 7\lambda = 2 + 5\mu$ and $1 + \lambda = -4 - 2\mu$, so $\boxed{\lambda = -1, \mu = -2}$. The third: $-13 + 2(-1) = -15$ but $2 - (-2) = 4$. No common point and not parallel: $\boxed{\text{skew}}$.

9 The directions $(0, a, 1)$ and $(1, 2, 3)$ are perpendicular: $2a + 3 = 0$, $a = -\frac{3}{2}$. Equating components: $4 = 1 + \mu$ gives $\mu = 3$; $-\frac{3}{2}\lambda = 2\mu = 6$ gives $\lambda = -4$; $-1 + \lambda =$

$$-b + 3\mu \quad \text{gives} \quad -5 = -b + 9. \quad \text{So}$$

$$\boxed{a = -\frac{3}{2}, b = 14}.$$

$$\mathbf{10 (a)} \quad \overrightarrow{AB} = \begin{pmatrix} k-1 \\ -4 \\ -2 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix}$$

$$\mathbf{10 (b)} \quad \text{On one line, } \overrightarrow{AB} = 2\overrightarrow{AC}: k-1=8, \quad \boxed{k=9}.$$

$$\mathbf{10 (c)(i)} \quad \boxed{\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix}}$$

$$\mathbf{10 (c)(ii)} \quad L_2: x = 1 + 2\mu, y = 3\mu, z = 1 - \mu.$$

The directions $(4, -2, -1)$ and $(2, 3, -1)$ are not multiples. $1 + 4\lambda = 1 + 2\mu$ and $2 - 2\lambda =$

$$3\mu \text{ give } \boxed{\lambda = \frac{1}{4}, \mu = \frac{1}{2}}, \text{ and the third fails:}$$

$$3 - \frac{1}{4} = 2.75 \neq 1 - \frac{1}{2} = 0.5. \quad \boxed{\text{skew}}$$

$$\mathbf{11 (a)} \quad B \text{ moves } 4 \text{ east and } 2 \text{ north: } \arctan \frac{4}{2} = 63.4^\circ \text{ from north, } \boxed{063^\circ}.$$

11 (b) $|(-6, 2, 4)| = \sqrt{56} = \boxed{7.48}$ km/min
and $|(4, 2, -2)| = \sqrt{24} = \boxed{4.90}$ km/min.

11 (c) $\cos \theta = \frac{-24+4-8}{\sqrt{56}\sqrt{24}} = -0.7637\dots$, $\theta = 139.8^\circ$, and the acute angle is $180^\circ - 139.8^\circ = \boxed{40.2^\circ}$.

11 (d)(i) $19 - 6t_1 = 1 + 4t_2$, $-1 + 2t_1 = 2t_2$,
 $1 + 4t_1 = 12 - 2t_2$: $t_1 = 2$, $t_2 = \frac{3}{2}$, and $P = \boxed{(7, 3, 9)}$.

11 (d)(ii) $|t_1 - t_2| = \boxed{0.5}$ minutes — 30 seconds.

12 (a) The velocity is $4(10, -25, 0) = (40, -100, 0)$: speed $\sqrt{11600} = 20\sqrt{29} = 107.703\dots = \boxed{108}$ km/h.

12 (b) $\tan \beta = \frac{16}{20\sqrt{29}}$, so $\beta = 8.44984\dots = \boxed{8.45^\circ}$.

Timer

$1 + 2s = 9 + 4t$ and $2 + 3s = 9 + t$ give $s = 2$, $t = -1$; the third component agrees, $-3 +$

$$12 = 9 = 11 - 2. \text{ So } \overrightarrow{OA} = \begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}.$$